Two-Phase Method

In the first phase of this method, the sum of the artificial variables is minimized subject to the given constraints in order to get a basic feasible solution of the LP problem. The second phase minimizes the original objective function starting with the basic feasible solution obtained at the end of the first phase. Since the solution of the LP problem is completed in two phases, this method is called the two-phase method.

Advantages of the method

1. No assumptions on the original system of constraints are made, i.e. the system may be redundant,
inconsistent or not solvable in non-negative numbers.
2. It is easy to obtain an initial basic feasible solution for Phase I.
3. The basic feasible solution (if it exists) obtained at the end of phase I is used as initial solution for Phase II.

Use two-phase simplex method to solve the following LP problem:

$$min \quad Z = 6x_1 + 21x_2$$

subject to the constraints

$$x_1 + 2x_2 \geq 3$$

$$x_1 + 4x_2 \geq 4$$

$$x_1, x_2 \geq 0$$

Converting the given LP problem objective function into the maximization form and then adding surplus variables S_1 and S_2 and artificial variables A_1 and A_2 in the constraints, the problem becomes:

$$max \quad Z^* = -6x_1 - 21x_2$$

Such that

subject to the constraints

$$x_1 + 2x_2 - S_1 + A_1 = 3$$

$$x_1 + 4x_2 - S_2 + A_2 = 4$$

$x_1, x_2, S_1, S_2, A_1, A_2 \geq 0$,where

$Z^* = -Z$

Phase I: This phase starts by considering the following auxiliary LP problem:

$$max \quad Z^* = -A_1 - A_2$$

Such that

subject to the constraints

$$x_1 + 2x_2 - S_1 + A_1 = 3$$

$$x_1 + 4x_2 - S_2 + A_2 = 4$$

$x_1, x_2, S_1, S_2, A_1, A_2 \geq 0$,where
$Z^* = -Z$

The initial solution

	C_j	0	0	0	0	-1	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	A_2	X_B
-1	A_1	1	2	-1	0	1	0	3
-1	A_2	1	4	0	-1	0	1	4
	Z_j	-2	-6	1	1	-1	-1	
	Z_j-C_j	-2	-6	1	1	0	0	

x_2 is entering basis

The initial solution

	C_j	0	0	0	0	-1	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	A_2	X_B
-1	A_1	1	2	-1	0	1	0	$\frac{3}{2}$
-1	A_2	1	4	0	-1	0	1	$\frac{4}{4}$
	Z_j							
	Z_j-C_j							

A_2 is leaving basis

x_2 is entering basis

The initial solution

	C_j	0	0	0	0	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	X_B
-1	A_1	1	2	-1	0	1	3
0	x_2	1	4	0	-1	0	4
	Z_j						
	Z_j-C_j						

The initial solution

	C_j	0	0	0	0	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	X_B
-1	A_1	1	2	-1	0	1	3
0	x_2	1	4	0	-1	0	4
	Z_j						
	Z_j-C_j						

$$R'_1 = R_1 - \frac{1}{2}R_2, R'_2 = \frac{1}{4}R_2 \ ,$$

The initial solution

	C_j	0	0	0	0	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	X_B
-1	A_1	$\frac{1}{2}$	0	-1	$\frac{1}{2}$	1	1
0	x_2	$\frac{1}{4}$	1	0	$-\frac{1}{4}$	0	1
	Z_j	$-\frac{1}{2}$	0	1	$-\frac{1}{2}$	-1	
	Z_j-C_j	$-\frac{1}{2}$	0	1	$-\frac{1}{2}$	0	

S_2 is entering basis

The initial solution

	C_j	0	0	0	0	-1		
C_B	B_V	x_1	x_2	S_1	S_2	A_1	X_B	$\frac{X_B}{x_j}$
-1	A_1	$\frac{1}{2}$	0	-1	$\frac{1}{2}$	1	1	$\frac{1}{1/2}$
0	x_2	$\frac{1}{4}$	1	0	$-\frac{1}{4}$	0	1	$\frac{1}{-1/4}$
	Z_j							
	Z_j-C_j							

A_1 is leaving basis

S_2 is entering basis

The initial solution

	C_j	0	0	0	0	-1	
C_B	B_V	x_1	x_2	S_1	S_2	A_1	X_B
0	S_2	$\frac{1}{2}$	0	-1	$\frac{1}{2}$	1	1
0	x_2	$\frac{1}{4}$	1	0	$-\frac{1}{4}$	0	1
	Z_j						
	Z_j-C_j						

$\boldsymbol{A_1}$ is leaving basis

$\boldsymbol{S_2}$ is entering basis

The initial solution

	C_j	0	0	0	0	
C_B	B_V	x_1	x_2	S_1	S_2	X_B
0	S_2	$\frac{1}{2}$	0	-1	$\frac{1}{2}$	1
0	x_2	$\frac{1}{4}$	1	0	$-\frac{1}{4}$	1
	Z_j					
	Z_j-C_j					

$$R'_1 = 2R_1, \;\; R'_2 = R_2 + \tfrac{1}{2}R_1,$$

The initial solution

	C_j	0	0	0	0	
C_B	B_V	x_1	x_2	S_1	S_2	X_B
0	S_2	1	0	-2	1	2
0	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{3}{2}$
	Z_j	0	0	0	0	
	Z_j-C_j	0	0	0	0	

Since all $\boldsymbol{Z_j}$-$\boldsymbol{C_j} \geq \mathbf{0}$ correspond to non-basic variables, the optimal solution: $x_1 = 0, x_2 = 0, S_1 = 0, S_2 = 0, A_1 = 0, A_2 = 0$ with $Z^* = 0$ is arrived at. However, this solution may or may not be the basic feasible solution to the original LP problem. Thus, we have to move to Phase II to get an optimal solution to our original LP problem.

Phase II: The modified simplex table

	C_j	-6	-21	0	0	
C_B	B_V	x_1	x_2	S_1	S_2	X_B
0	S_2	1	0	-2	1	2
-21	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{3}{2}$
	Z_j	$\frac{-21}{2}$	-21	$\frac{21}{2}$	0	
	Z_j-C_j	$\frac{-9}{2}$	0	$\frac{21}{2}$	0	

x_1 is entering basis

Phase II: The modified simplex table

	C_j	-6	-21	0	0		
C_B	B_V	x_1	x_2	S_1	S_2	X_B	$\frac{X_B}{x_j}$
0	S_2	1	0	-2	1	2	$\frac{X_2}{1}$
-21	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	0	$\frac{3}{2}$	$\frac{3/2}{0}$
	Z_j						
	Z_j-C_j						

S_2 is leaving basis

x_1 is entering basis

	C_j	-6	-21	0	
C_B	B_V	x_1	x_2	S_1	X_B
-6	x_1	1	0	-2	2
-21	x_2	$\frac{1}{2}$	1	$-\frac{1}{2}$	$\frac{3}{2}$
	Z_j				
	Z_j-C_j				

$$R'_1 = R_1, \quad R'_2 = R_2 - \frac{1}{2}R_1,$$

	C_j	-6	-21	0	
C_B	B_V	x_1	x_2	S_1	X_B
-6	x_1	1	0	-2	2
-21	x_2	0	1	$\frac{1}{2}$	$\frac{1}{2}$
	Z_j	-6	-21	$\frac{3}{2}$	
	Z_j-C_j	0	0	$\frac{3}{2}$	

$\boldsymbol{Z_j\text{-}C_j \geq 0}$ the optimal solution: $x_1 = 2, x_2 = \frac{1}{2}, S_1 = 0, S_2 = 0, A_1 = 0, A_2 = 0$ with $Z^* = \frac{-45}{2}$, $\min Z = \frac{45}{2}$